

REPORT CONTENT SHEET

Raw material for your futuristic report — Build It in English

Hoja de contenido: materia prima del informe futurista

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Project / Proyecto	HABIT LOOP — Habit Tracker
Pitch	A lightweight PHP web application to build, track, and maintain daily habits effortlessly.
Environment / Entorno	PHP 8.x + Apache (XAMPP)
Report medium / Medio del informe	Word exported to PDF
Date / Fecha	2026-09-01
Teacher / Docente	Ing. Javier Arturo García Mineros

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2. The environment used

PHP is lightweight, easy to deploy locally with XAMPP, and native to server-side script execution. It allows simple file handling (JSON) or database queries (MySQL) without complex configuration overhead, making it perfect for rapid habit tracking logic.

PHP es ligero, fácil de ejecutar localmente con XAMPP y nativo para scripts de servidor. Permite gestionar archivos JSON o bases de datos MySQL de forma sencilla y sin configuraciones complejas, lo que lo hace ideal para desarrollar la lógica del rastreador de hábitos rápidamente.

2.1 Alternative not used

Node.js was considered, but skipped because it requires managing NPM packages and asynchronous runtime setups that add unnecessary complexity for a simple habit tracking project.

Se consideró Node.js, pero se descartó porque requiere administrar paquetes de NPM y configuraciones asíncronas que añaden complejidad innecesaria para un proyecto simple de seguimiento de hábitos.

3. Tools

TOOL	VERSION	WHAT IT WAS USED FOR (ENGLISH)
Visual Studio Code	1.85.0	Writing and editing backend PHP code, HTML structure, and CSS styling.
XAMPP	8.2.12	Running the local Apache web server and PHP execution environment.
Visual Studio Code	Latest	Testing application responsiveness, UI layout, and browser debugging.
Git	2.43.0	Managing source code version control and tracking project changes.

4. Resources

RESOURCE TYPE	DETAIL	CONTRIBUTION TO THE PROJECT
Hardware	Laptop Intel Core i5, 8GB RAM	Provided the primary development environment for writing code and testing locally.
Time invested	12 hours	Divided between Sprint 0 charter preparation, backend logic, and UI design.
Documentation / tutorials	PHP Official Documentation & W3Schools	Consulted for syntax validation, date handling methods, and file system logic.
Libraries or packages	Bootstrap 5 / FontAwesome	Provided basic layout styling components and UI icons for habit status.

AI or assistance used	ChatGPT / Gemini	Assisted in structuring technical documentation, drafting English microcopy, and code debugging.
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5. Functionality

FEATURE	WHAT THE USER DOES AND SEES (ENGLISH)	EVIDENCE
Habit Creation	The user clicks "+ Add New Habit", fills in the name and weekly target, and sees the new habit listed on the main dashboard.	
Daily Check-in	The user clicks the checkbox next to a habit. The status updates instantly to completed and increments the streak counter.	
Progress Tracking	The user opens the progress view and sees a weekly percentage chart reflecting their completed target days.	
Habit Deletion	The user clicks the delete button on a habit, confirms the dialog, and sees the habit removed from the list.	

5.1 Out of scope

The application does not support multi-user authentication, cloud synchronization across devices, push notifications outside the browser session, or social sharing features.

La aplicación no incluye autenticación de múltiples usuarios, sincronización en la nube entre diferentes dispositivos, notificaciones push fuera del navegador, ni funciones para compartir en redes sociales.

6. Architecture and data model

The application follows a simple Modular MVC structure. The `/public` directory holds entry points (`index.php`) and static assets (`/css`, `/js`). Core application logic resides in `/src`, featuring `Habit.php` for data models and `HabitController.php` for handling actions like adding or completing habits. The `/config` folder contains database setup scripts and global helper functions.

La aplicación sigue una estructura Modular MVC sencilla. El directorio `/public` contiene el punto de entrada (`index.php`) y archivos estáticos (`/css`, `/js`). La lógica principal está en `/src`, con `Habit.php` para los modelos de datos y `HabitController.php` para gestionar acciones como agregar o completar hábitos. La carpeta `/config` incluye la configuración de la base de datos y funciones globales.

6.1 Data persistence

Data persists locally inside a MySQL database managed via PHP Data Objects (PDO) for safe queries, with a secondary JSON file backup stored in /data/habits.json. Each user action updates the database instantly, ensuring full data persistence across browser sessions.

Los datos se persisten localmente en una base de datos MySQL administrada mediante Objetos de Datos de PHP (PDO) para consultas seguras, junto con un respaldo secundario en un archivo JSON guardado en /data/habits.json. Cada acción del usuario actualiza la base de datos al instante, garantizando la persistencia total entre sesiones del navegador.

Insert here the diagram of your data model.
Inserte aquí el diagrama de su modelo de datos.

7. Code highlights

FILE AND METHOD	WHAT IT DOES	WHY IT IS INTERESTING
HabitController::calculateStreak()	Iterates through historical logs to count consecutive active days.	Uses PHP DateTime objects to accurately calculate streaks across different months and leap years without date bugs.
Database::getConnection()	Initializes a Singleton connection to MySQL via PDO.	Prevents multiple open connections by reusing a single database instance, optimizing performance and server memory.
Habit::saveToJson()	Encodes current habit data arrays into structured JSON text.	Implements a simple file-lock mechanism to ensure data integrity during simultaneous write operations.

Paste the three code excerpts below, with their English comments.
Pegue los tres fragmentos de código con sus comentarios en inglés.

8. Test log

ID	WHAT WAS TESTED	INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
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T-01	Create new habit with valid data	Name: "Drink Water", Target: 5	Habit is saved and displayed on dashboard	Habit appeared on list instantly	PASS
T-02	Submit habit form with empty name	Name: "", Target: 3	Validation error message displayed	Error "Please enter a habit name" shown	PASS
T-03	Daily check-in completion	Click check-box for "Drink Water"	Habit marked completed, streak increments by 1	Streak incremented from 0 to 1	PASS
T-04	Streak calculation across months	Check-in on Aug 31 and Sep 01	Racha updates to 2 consecutive days	Racha accurately calculated as 2 days	PASS
T-05	Delete an existing habit	Click delete on "Drink Water"	Habit is removed from UI and storage	Habit successfully removed	PASS
T-06	Dark/Light theme toggle switch	Toggle ThemeToggleSwitch	UI background and font colors shift	Theme switched dynamically	PASS
T-07	Data persistence test	Refresh browser page	Previously logged habits remain intact	All data reloaded correctly from storage	PASS
T-08	Weekly progress percentage calculation	3 out of 7 days checked	Progress bar shows ~43% completion	Bar filled to 43% with exact label	PASS

9. Challenges and conclusions

PROBLEM	WHAT I SEARCHED IN ENGLISH	HOW I SOLVED IT
Streak calculation bug across months	"PHP calculate consecutive days between dates across different months"	Used native PHP DateTime diff methods instead of simple string date comparisons.
JSON file write permission error	"XAMPP PHP JSON file write permission denied local server"	Updated local directory permissions and used file_put_contents with LOCK_EX.

9.1 Conclusions

Developing Habit Loop reinforced the importance of clear modular code structure and accurate date handling in tracking software. We learned how to build a clean PHP backend combined with light frontend interactions. Next time, we would implement a database migration script early on and plan unit tests from the start to catch edge cases faster.

El desarrollo de Habit Loop reforzó la importancia de mantener una estructura de código modular clara y de manejar las fechas con precisión en aplicaciones de seguimiento. Aprendimos a construir un backend eficiente en PHP combinado con interacciones ligeras en el frontend. Para la próxima ocasión, implementaríamos un script de migración de base de datos desde el principio y planificaríamos pruebas unitarias tempranas para detectar errores de borde más rápido.

Declaration / Declaración

I declare that I wrote the code and the report myself, that every identifier and comment is in English, and that I can explain any part of this work on request.

Declaro que escribí el código y el informe, que todo identificador y comentario está en inglés y que puedo explicar cualquier parte de este trabajo.

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